

PLSC 197 / SODA 110N: Social Data, Technology, and Artificial Intelligence

Assignment 4: Learning Paradigms

Assignment

In line with our week of talking about different kinds of learning models, this assignment focuses on *learning paradigms*. It's pretty straightforward:

1. First, choose a real-world AI application, ideally one you know and/or use. Suggestions include

- Amazon's recommendations
- Apple/Google Maps' routemaker
- Shazam's song finder
- Your bank's fraud detector
- Grubhub
- Spotify's "More Of What You Like"
- Google News' aggregator
- Instagram's post-orderer
- Bumble/Hinge/etc.
- Image generators (Sora 2, Dall-E, etc.)
- DuoLingo / Speak / etc.
- Your email's spam filter
- Uber

2. Write a short brief containing the following:

- (a) Explain whether the application uses supervised, unsupervised, or reinforcement learning, and how you know.
- (b) What kind of data does it require / use?
- (c) Indicate if it uses / is capable of deep, transfer, and/or one-shot learning.
- (d) What kinds of errors is it likely to make, and why?
- (e) Speculate briefly on what changes to its learning model might improve the application's performance.

Details

Your assignment should include a separate title page with your name, the name of the course, the name of the assignment, and the date on which it was completed / submitted. The body of the written assignment should be *double-spaced*, in a conventionally-sized (11 or 12 point) font. Each written part of the assignment should be as short or long as you deem necessary; a rough target might be a total of 3-4 pages. You need not cite any sources for this assignment, or include references, unless you choose to.

This assignment must be completed and submitted by Friday, October 16, 2026 at 11:59 p.m. ET. Submit your assignment by *creating / converting your assignment into PDF format*, and emailing it to Dr. Zorn (at zorn@psu.edu). Assignments received after the due date will not be accepted.

This assignment is worth a possible 100 points.